



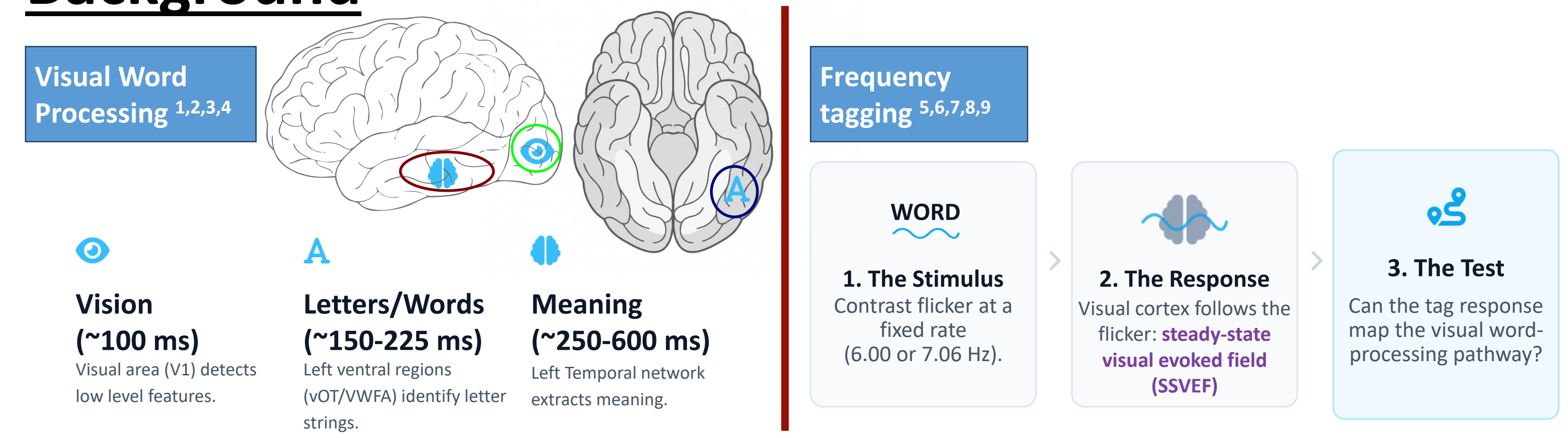
Characterizing word processing pathway using frequency tagging

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A **6 / 7.06 Hz flicker tag** tracks the visual input, not wordness; the **word-locked response** (250–600 ms) separates words from consonant strings, and in anterior temporal cortex the difference is sustained across words.

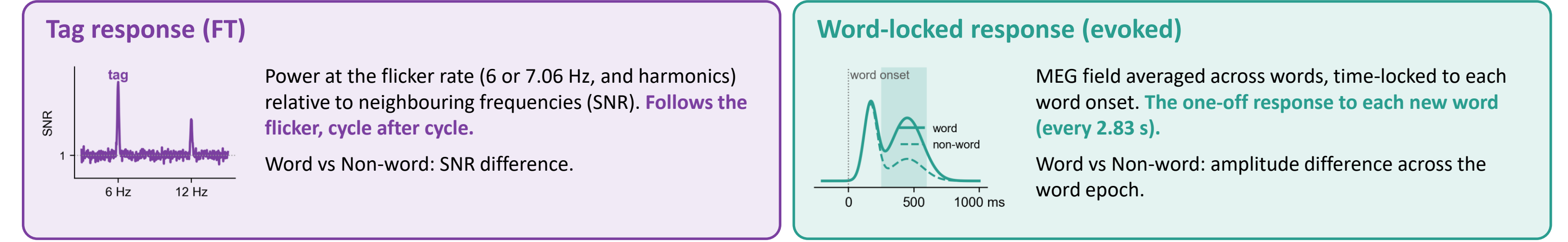
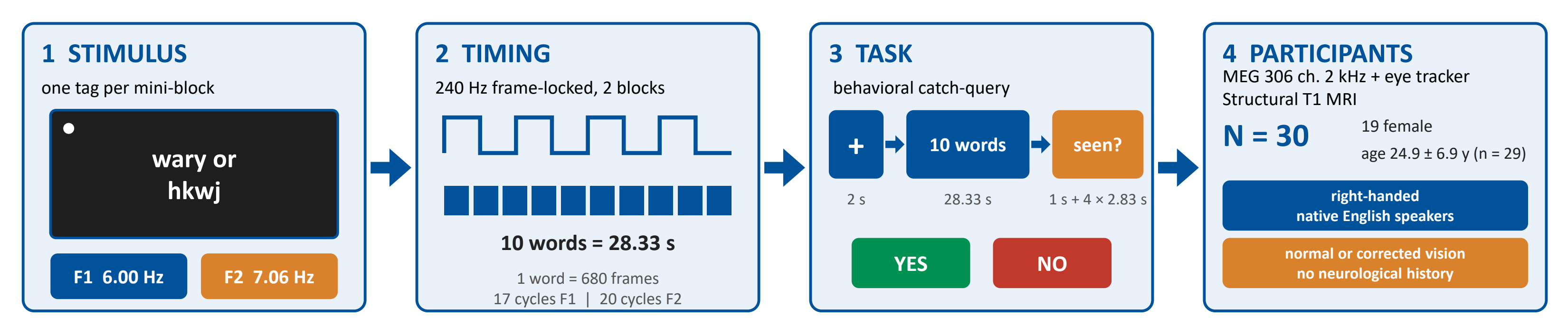
Update vs. submitted abstract: with fixation baselines and equal trial counts, the tag shows no word-specific effect and the source effect is Word > Non-word.

Background



- AIM 1**
Do MEG sensors and sources recover the V1 → vOT → temporal sequence of the literature?
- AIM 2**
Does the tag response (SSVEF) differ between words and consonant strings?

Methods



Pitfalls we hit — and how we fixed them

- Tags off by one FFT bin.** The display ran ~0.5 % slow (photodiode: 5.965 / 7.024 Hz): read the SNR at the measured bin.
- Unequal trial counts faked an effect.** A time-resolved source effect vanished once counts were equalized (the whole-epoch effect remains).
- No gap between words.** The pre-stimulus window holds the previous word: baseline on fixation instead.
- Check the noise-covariance rank.** After SSS, 3 subjects kept a ~1e-38 eigenvalue (dSPM × 10⁶, distorted maps): fixed with `mne.compute_rank(tol='auto')`.



Code · supplementary figures S1–S4 · cohort
github.com/Orion212-lab/snl2026-ft-word-pathway

Cohort 30 recorded; each panel states its N. Excluded: no flicker on the photodiode, malformed blocks, all epochs rejected; sources: no T1 MRI, artifact (sub-20/24). Kept with trigger anomalies: sub-10–14 (main results unchanged without them). List via QR.

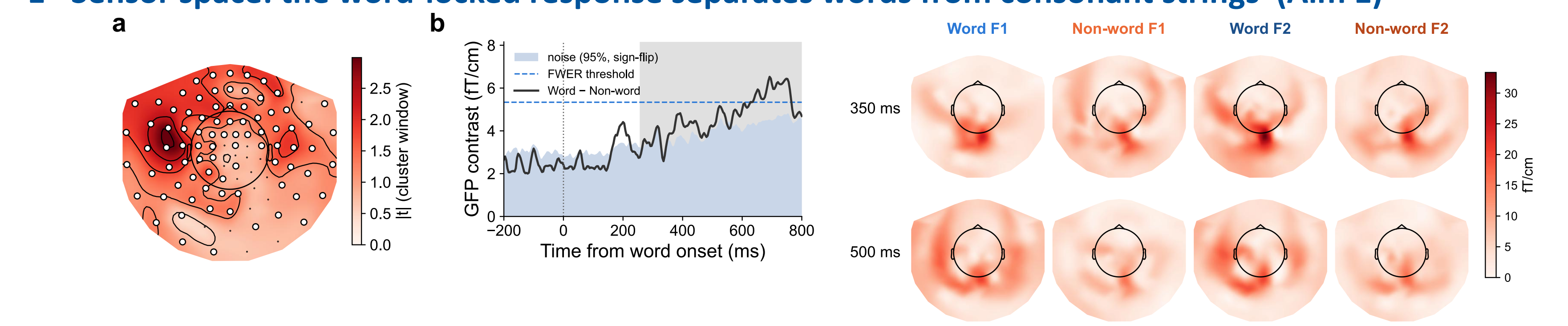
A priori Word vs Non-word at the tag and in the evoked cascade windows (80–130, 150–225, 250–600 ms), left-temporal sensors. Not preregistered; everything else is exploratory.

Limits consonant strings also differ in orthography, pronounceability, visual complexity and task difficulty · condition-pure blocks · late vOT effect fragile · catch-query accuracy not yet scored

Next pseudowords · item-rate (FPVS) tagging instead of contrast flicker · interleaved conditions · two-word phrase task (intermodulation), recorded, not yet analysed

Results

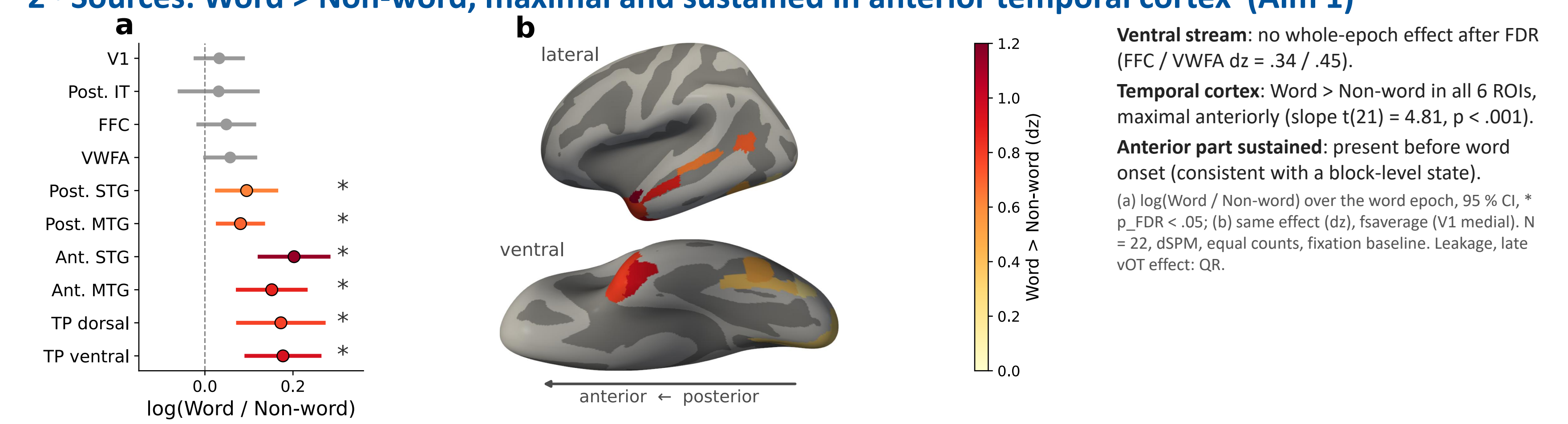
1 · Sensor space: the word-locked response separates words from consonant strings (Aim 1)



One Word > Non-word cluster over left posterior and temporal sensors (N = 26, 84 sensors, $p < .001$), robust to blinks, saccades and artifacts (clean epochs: $dz = 1.31$, 20/21 subjects). Tag × Wordness: n.s.

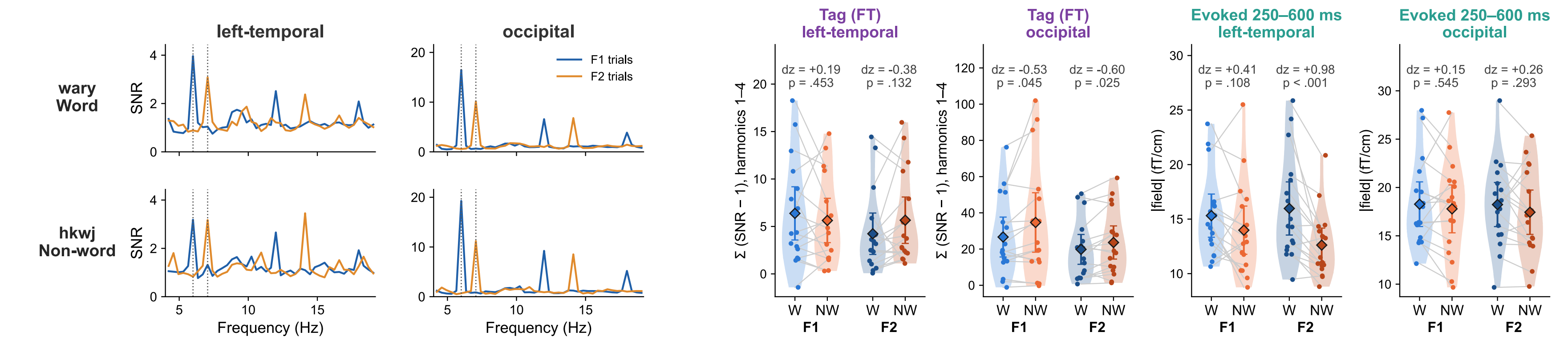
(a) mean $|t|$ in the cluster window (N = 26); (b) GFP of W - NW vs sign-flip noise (shaded: 95 %; dashed: FWER), exceeded at 628–764 ms; right: grand averages (N = 25) at 350 / 500 ms, ±25 ms; fixation baseline.

2 · Sources: Word > Non-word, maximal and sustained in anterior temporal cortex (Aim 1)



Ventral stream: no whole-epoch effect after FDR (FFC / VWFA $dz = .34 / .45$).
Temporal cortex: Word > Non-word in all 6 ROIs, maximal anteriorly (slope $t(21) = 4.81$, $p < .001$).
Anterior part sustained: present before word onset (consistent with a block-level state).
(a) $\log(\text{Word} / \text{Non-word})$ over the word epoch, 95 % CI, * $p_{\text{FDR}} < .05$; (b) same effect (dz), fsaverage (V1 medial). N = 22, dSPM, equal counts, fixation baseline. Leakage, late vOT effect: QR.

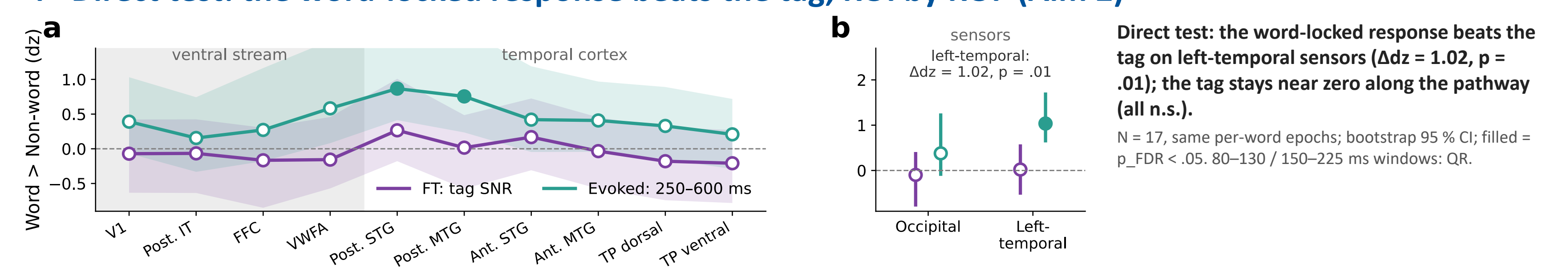
3 · The tag follows the visual input, not wordness (Aim 2)



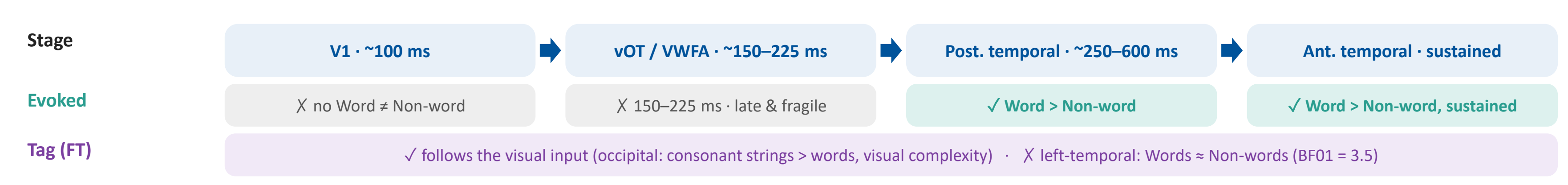
Tag (harmonics 1–4): occipital Non-words > Words ($dz = -.78$, $p = .006$), matching their extra visual complexity (+10 % ink); left-temporal Words = Non-words ($dz = -.14$, $BF01 = 3.5$). **Evoked** 250–600 ms, the reverse: left-temporal Words > Non-words ($dz = .93$, $p = .002$; Tag × Wordness n.s.), occipital n.s.

Same 17 subjects. Left: per-word phase-locked SNR (2.83 s FFT). Right: dots = subjects, diamonds = mean ± 95 % CI; dz / p per tag (paired); combined effects in the text.

4 · Direct test: the word-locked response beats the tag, ROI by ROI (Aim 2)



Direct test: the word-locked response beats the tag on left-temporal sensors ($\Delta dz = 1.02$, $p = .01$); the tag stays near zero along the pathway (all n.s.).
N = 17, same per-word epochs; bootstrap 95 % CI; filled = $p_{\text{FDR}} < .05$. 80–130 / 150–225 ms windows: QR.



Aim 1 — partly: temporal stages replicated, vOT only late and fragile, latency order n.s. **Aim 2 — no:** the tag indexes visual input, not wordness; the word-locked response does.

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